MIL-P-36477A
19 February 1976
SUPERSEDING
MIL-P-36477
12 April 1966

MILITARY SPECIFICATION

PROBES, PERIODONTAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers probes suitable for use in measuring the depth of periodontal pockets.

1.2 Classification. Probes shall be of the following styles, as specified (see 6.1);

Style 1. Fox.
Style 2. Merritt "B".

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

FSC 6520

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Military

MIL-P-116	Methods of Preservation.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof Or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-66	Steel, Chemical Composition and Hardenability.
FED-STD-151	Metal, Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center (DPSC)

23027	Probes, Periodontal Fox and Merritt "B"

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 135	Specification for Seamless Brass Tube.
E 18	Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
E 92	Standard Method of Test for Vickers Hardness of Metallic Materials.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 General requirements. The probes shall be in accordance with the applicable requirements of GG-I-526, and as hereinafter specified.

3.2 Material.

3.2.1 Operative ends. The operative ends shall be Class 3, Type 302, 303, or 304 stainless steel.

3.2.2 Handles. The handles shall be Class 3 Type 303 or 416 stainless steel, or Class 6 stainless steel except the type shall be 430F of FED-STD-66 in lieu of type 430, or Class 8 Alloy Number 360 free cutting brass, chromium or nickel plated. As an alternate to Class 8 Alloy Number 360 composition, the brass composition only shall be in accordance with Alloy Number 330 of ASTM B 135. The boil test shall apply for the steel and brass handles. The copper sulfate test shall apply for handles made of Class 6 steel.

3.3 Hardness. After proper stress relieving, the Rockwell hardness of the operative ends shall be C45 to C50.

3.4 Style, design and dimensions. Style 1 shall be a Fox design probe with a flat blade and Style 2 shall be a Merritt "B" design probe with a round blade. The probes shall be marked off in millimeters and shall be suitable for use in measuring the depth of periodontal pockets. The style, design, and dimensions of the probes shall conform to Defense Personnel Support Center (DPSC) Drawing 23027, for the applicable style.

3.4.1 Drawings. The drawings forming a part of this specification are specification control drawings. The supplier shall be responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper design and function of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

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3.5 Construction. The probes shall consist of an operative end and a handle. The operative ends shall be permanently secured in the handles. The millimeter markings shall be legible sharply defined lines, of uniform width and depth. The finger grip portion of the handles shall be finely serrated or fine diamond knurled.

3.6 Finish. The operative ends shall have a mirror finish. The handles shall have a mirror or satin finish, except for the serrated or knurled portion. The operative ends shall be free from coarse grind marks, cracks, burrs, feathers, or other defects when viewed as specified in 4.3.1.

3.7 Identification marking. Style 1 probe shall be marked with the identification marking "Fox" and style 2 probe shall be marked with "Merritt B." Marking shall be on the front face of the instruments. All markings shall be on the smooth portion of the handles and shall be permanent and legible. As an alternate to the word "stainless," the use of a registered trade name identifying the steel as stainless is acceptable, provided prior approval is received from the cognizant contracting officer (see 6.1).

3.8 Workmanship. Workmanship shall be first class throughout. The probes shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in Table I.

Table I. Examination criteria

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-1	2.5
Minor defects	II	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as specified in Table II, except that the acceptance number shall be zero for all sample sizes. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics. Lots for hardness testing shall be as specified in 4.2.2.2.

Table II. Test criteria

Component	Characteristics	Requirement	Test procedure	Inspection level
Operative end	Hardness	3.3 and GG-I-526	4.4.2	S-2
Handle (430F steel only)	Copper Sulfate	3.2.2 and GG-I-526	GG-I-526	S-2
Entire probe	Boil test	3.2.1, 3.2.2 and GG-I-526	GG-I-526	S-2
Entire probe	Composition	3.2.1 & 3.2.2	4.4.1	See 4.2.2.1

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4.2.2.1 For raw material. Sufficient amount from each lot of each metal used in the construction of the probes shall be selected for composition analysis.

4.2.2.2 Lots for hardness criteria. For finished instrument (operative end) hardness testing, a lot or batch of instruments shall consist of instruments (operative ends) produced by one manufacturer at one plant, from the same material (same raw material heat and heat treatment) and from the same heat treatment lot or batch (instruments undergoing heat treatment at the same time and in the same heat treatment equipment).

4.3 Examination. The probes shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with the classification of defects reflected below in Table III. Any probes in the sample containing one or more defects shall be rejected, and if the number of defective probes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table III. Classification of defects

Categories	Defects 1/
Critical	None defined.
Major A:	
101	Style and design not as specified for the applicable style.
102	Instrument not constructed as specified.
103	Operative end finish not as specified.
104	Operative end not free from cracks and burrs.
Major B:	
151	Operative end dimensions not as specified.
Minor:	
201	Handle finish not as specified.
202	Instrument not permanently and legibly marked as specified.
203	Instrument identification not properly located.
204	Handle dimensions not as specified.

1/ Examination shall not be restricted to the classified possible defects listed.

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4.3.1 Microscopic examination. Examination of probe defects not readily perceptible to the naked eye shall be made using a microscope at 4 to 10 power (4X to 10X) magnification.

4.3.2 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical analysis. In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with method 111 or method 112 of Federal Test Method Standard No. 151. In case of dispute, method 111 shall govern.

4.4.2 Hardness. Hardness shall be determined in accordance with ASTM Designation E 18 (Rockwell) or E 92 (Vickers or Diamond Pyramid).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Each probe shall be thoroughly cleaned and dried in accordance with the applicable method in MIL-P-116. Each probe shall be wrapped in neutral tissue or other non-corrosive material and packaged in accordance with MIL-P-116, method IC-2. A preservative shall not be applied to the probe.

5.1.1.2 Intermediate package. Twelve unit packages containing probes of one style only shall be packaged in a box constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure of the box shall be as specified in the appendix of the applicable box specification.

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5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Twenty four intermediate packages (288 units) containing probes of one style only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601 overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Twenty four intermediate packages (288 units) containing probes of one style only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

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5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Style required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Approval of trade name, if applicable (see 3.7).

6.2 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with asterisks to indicate where changes from the previous issue have been made.

6.3 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

NSN	Item Identification
6520-00-890-2048	PROBE, PERIODONTAL, Fox
6520-00-890-2049	PROBE, PERIODONTAL, Merritt "B"

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1400

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-P-36477A PROBES, PERIODONTAL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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